Travis Pan

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Education

University of Waterloo — Waterloo, Ontario
Bachelor of Computer Science and Finance (Double Major) — *Sep. 2023 – Apr. 2028 (Expected)*

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, Kotlin, C++, SQL
Libraries & Frameworks: React, Next.js, Node.js, Tailwind, Spring Boot, FastAPI, Django
Technologies: Docker, AWS, Kubernetes, Linux, Apache Spark, Apache Iceberg, Kafka, Flink, PostgreSQL, Git
Certifications: AWS Certified Cloud Practitioner

Experience

Incoming Software Engineer @ Rippling — Winter 2027 — San Francisco, California

Software Engineer - Data Platform — May 2026 – Aug. 2026
Faire — *Toronto, Ontario*

- Built a high-throughput **Kafka** and **Protobuf** event platform supporting hundreds of MB/s, reducing network bandwidth by **82%** (560 MB/s to 100 MB/s) through schema-registered Protobuf
- Built **Kotlin/Spring Boot** backend services that emitted API request telemetry to **Kafka** for real-time latency, error-rate, and anomaly monitoring with **Flink**, processing **1M+ daily events** at **˜50 ms p95 latency**
- Migrated production batch jobs to **AWS Graviton** with multi-architecture ARM64/AMD64 **Docker** images and **Kubernetes**/**Terraform** infrastructure, reducing compute costs by $10K/month
- Built a **Python** benchmarking framework comparing cost and performance across **Snowflake**, **Databricks**, and **DuckDB**, adopted across the ML platform team to identify lower-cost execution strategies

Software Engineer — Sep. 2025 – Apr. 2026
AlphaSense — *New York, New York*

- Built a custom **SQL** compiler using **AST transformations** to compile analyst-defined business rules into optimized **Spark** queries, backed by a **Django**-based rules engine as the interface
- Built **FastAPI** microservices over **Apache Iceberg** tables in **S3**, integrating **AWS SQS** event-driven pipelines with **Redis**-backed state to enable asynchronous data processing
- Built and optimized distributed data pipelines using **Spark SQL** and **Apache Iceberg** on **Kubernetes**, reducing latency by 30% through query and partitioning optimizations
- Designed and built a high-performance internal rules editor with **React** frontend and **Django** backend, supporting session-based state, delta comparison, and rule validation for low-latency interactions over large-scale datasets

Software Engineer — Jan. 2025 – Apr. 2025
Government of Canada - PSPC — *Toronto, Ontario*

- Built a full-stack AI tooling web application using **Next.js** and the **OpenAI API**, enabling non-technical users to run RAG workflows and agents that execute **SQL** queries and external APIs via **MCP**
- Built and deployed machine learning pipelines using **Azure ML**, integrating **SARIMA**, **XGBoost**, and **PyTorch LSTM** models into production data workflows with automated retraining

Projects

Codebase Quizlet | *FastAPI, PostgreSQL, Redis, SCIP*

- Built a repository analysis platform that uses **SCIP** indexing, dependency graphs, and Louvain clustering to model large codebases and generate structured knowledge representations for downstream applications
- Designed a distributed processing pipeline with **FastAPI**, **PostgreSQL**, **Redis**, asynchronous workers, and DAG orchestration to support scalable repository ingestion, analysis, and content generation

LeetDance (Hack the North 2025) | *Next.js, FastAPI, MediaPipe, OpenCV, Gemini API*

- Built an AI dance choreographer that reviews user performance by extracting skeletal joint positions with **MediaPipe**, processing movement accuracy via **OpenCV**, and generating real-time scores and visual overlays
- Designed an end-to-end pipeline where uploaded footage is streamed through a **FastAPI backend**, analyzed with **Gemini API** for annotated feedback, and delivered to a **Next.js frontend** as interactive insights